

Analytical Question:-

1. Formula:

Encryption: $C = (P + K) \bmod 26$

Decryption: $P = (C - K) \bmod 26$

K - key (15)

Encryption,

plaintext	H	E	L	L	O
position	7	4	11	11	14
calculation	$7+15$	$4+15$	$11+15$	$11+15$	$14+15$
Mod 26	22	19	0	0	3

Cipher text = WTAAD

decryption,

cipher	:	W	T	A	A	O
position	:	22	19	0	0	3
calculation	:	$22-15$	$19-15$	$0-15$	$0-15$	$3-15$
Mod 26	:	7	4	11	11	14

Decrypted: HELLO

2.) plaintext: MEET ME AFTER THE PARTY

Plain: MEET ME AFTER THE PARTY

cipher: PHHW PH DIWPU WKH SDUWB

Encrypted text:

PHHW PH DIWPU WKH SDUWB

decrypted text:

MEET ME AFTER THE PARTY.

3.)

key = 3 (A=1)

plain text: CRYPTOGRAPHY

letters: C R Y P T O G R A P H Y

Shift+3: F V B S W R J U D S K B

Encrypted text: FVBSWRJUDSKB

4.)

CRYPTOGRAPHY.

standard substitution table.

Plain: C R Y P T O G R A P H Y

cipher: E K N I H Z G U K Q H Z N

Encryption:

EKNIHZGUKQHZN

decryption:

using the same table,

CRYPTOGRAPHY.

5)

Plain : MEET ME TOMORROW IN THE MORNING

Cipher : DTTZ DT ZGOGKKGV OF FZI TDGKFOU

Encrypted text,

DTTZ DT ZGOGKKGV OF FZI TDGKFOU

decryption text,

Reverse the substitution table.

Recovered text:

MEET ME TOMORROW IN THE MORNING.

6).

keyword : KEYWORD

K	E	Y	W	O
R	D	A	B	C
F	G	H	I	L
M	N	P	Q	S
T	U	V	X	Z

Plain text :

MEET ME TOMORROW

Create pairs : ME ET ME TO MO RR OW

new pairs : ME ET ME TO RX RO WX

Cipher text :

NKKUNKZK7FBTKBW

8)

Plain text: BALOON

PAIRS:

B	A
L	O
O	X
O	N

Cipher: C B M Z P V

Decryption: BALOON.

9) Vigenere cipher:

plain text: ACTIVE

KEY: PASCAL

Repeat Key:

ACTIVE

PASCAL

Cipher: P U V K U P

decryption: ACTIVE

10) Vigenere cipher:

plain: SHE IS LISTENING-

key: P A S C A L

Repeat key over letters,

Cipher: ~~H~~ H W V S D I L T M N V C V

decryption: SHE I S LISTNENING.

11) - Hill cipher -

plain text : H E L P

Key matrix :

$$\begin{bmatrix} 3 & 3 \\ 2 & 5 \end{bmatrix}$$

pairs : H E
L P

cipher : H I A T

decryption : H E L P

12) - Hill cipher :-

plain text : T E X T

Same key matrix.

pairs :

T E
X T

Multiply each pair with key matrix.

Take mod 26

cipher text : R U M R

decryption : ~~T~~ T E X T

Dr